

The $1/3$ – $2/3$ Conjecture for Posets at Chain-Deletion Distance Two

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Abstract

A finite poset is a k -defect chain if it becomes a chain after the deletion of k elements. We prove the $1/3$ – $2/3$ conjecture for all two-defect chains: every such poset that is not itself a chain has an incomparable pair (u, v) with $1/3 \leq \Pr(u \prec v) \leq 2/3$. The class contains posets of width three and does not bound the number of incomparabilities per element, and we exhibit members escaping each previously proved class known to us. We prove equality rigidity: every ordinal-sum-indecomposable poset of defect exactly two has a strictly balanced eligible count, and among chain-plus-two-incomparable-defect presentations the unique non-strict member is the three-element poset Q (a two-element chain plus an isolated point, itself of defect one); this extends Aigner’s width-two equality classification to an infinite family of width-three posets, whose non-attainment of $1/3$ the order-14 census could previously only observe empirically. A closing section takes the first step onto the three-defect ladder: an infinite three-defect family attaining the boundary exactly, and the precise place where the two-defect arithmetic stops.

1 Introduction

The $1/3$ – $2/3$ conjecture — posed independently by Kislitsyn [1], Fredman [2], and Linial [3] in connection with sorting under partial information — asserts that every finite poset P that is not a chain contains an incomparable pair (u, v) whose orientation probability under the uniform distribution on linear extensions satisfies $1/3 \leq \Pr(u \prec v) \leq 2/3$. Kahn and Saks [4] proved the first uniform positive bound; the strongest general bound is due to Brightwell, Felsner, and Trotter [9]. The conjecture is known for partial orders of width two (Linial [3], sharpened by Sah [16]), height two [8], posets with at most 14 elements [12, 19], posets in which every element is incomparable to at most six others (Peczarski [10]), series-parallel posets, N -free posets and polytrees (Zaguia [13, 14]), semiorders (Brightwell [7]), several lattice and Young-diagram families (Olson and Sagan [15]), and is preserved by lexicographic sums [17]. Chan and Pak [18] survey the area.

Every finite poset becomes a chain after deleting some number of elements; call the least such number its *defect*. The conjecture restricted to defect k interpolates, as k grows, from the merging problems of the classical literature to the full conjecture. Defect one is contained in width two. This paper settles defect two completely, classifies its extremal cases, and closes with the first rung of the three-defect ladder.

Three subfamilies of the two-defect class were previously known and are not claimed here: members whose two defect windows coincide carry a defect-exchanging automorphism, a case

settled by Peczarski's automorphism theorem [11] (and, for cycle-free members, already by Ganter, Häfner, and Poguntke [6], who obtain a $1/2$ -balanced pair); members of width two are Linial–Sah; members whose windows have length at most two are semiorders. The class as a whole — with windows of unbounded length, hence unboundedly many incomparabilities at a single element, and with genuine width-three members — is contained in neither the width-two class nor any bounded-incomparability class, and its non-containment in the remaining listed classes is witnessed explicitly. With $m = 2$ and windows $I = [0, 1]$, $J = [1, 2]$, the four elements (x, c_1, c_2, y) induce an N ($x < c_2$, $c_1 < c_2$, $c_1 < y$, all other pairs incomparable), both as an induced suborder and in the Hasse diagram — so the class is contained in neither the series-parallel nor the N -free class. With $m = 3$ and windows $I = [1, 2]$, $J = [1, 3]$, the Hasse diagram contains the cycle $c_1 x c_3 c_2$ — so it is not contained in the forest-cover class. Unbounded windows exclude semiorders, and height excludes the height-two class. The incomparable-defect members are precisely the interval orders with exactly two nontrivial intervals; general interval orders remain open.

Section 2 constructs the insertion-window model and its state bijection; Section 3 proves the balance lemma; Section 4 assembles the Two-Defect Chain Theorem; Section 5 proves the equality rigidity classification; Section 6 poses the next problem. Throughout, $e(P)$ is the number of linear extensions of P , $\Pr(u \prec v)$ the fraction orienting u before v , and $\#L(P; \cdot)$ the number of extensions with the displayed property; windows are integer intervals of insertion gaps against the fixed chain C .

2 The doubled-diagonal model

Let $C = (c_1 < \dots < c_m)$ be a chain and let x, y be two further elements with $x \parallel y$. Consistency with the chain forces the admissible insertion gaps of x to form a window $I = [a, b] \subseteq \{0, \dots, m\}$ (gap i meaning that exactly i chain elements precede x): the relations are $c_t < x$ for $t \leq a$ and $x < c_t$ for $t > b$. Similarly y carries a window $J = [c, d]$.

Lemma 2.1 (Overlap). *If $x \parallel y$ then $I \cap J \neq \emptyset$. Conversely, for any two nonempty overlapping windows I, J , imposing the chain order, the threshold relations of both windows, and no relation between x and y defines a valid poset in which $x \parallel y$ and no hidden comparabilities arise.*

Proof. If $b < c$ then $x < c_{b+1} \leq c_c < y$ by transitivity, contradicting $x \parallel y$; the case $d < a$ is symmetric. For the converse, every pair of gaps $(i, j) \in I \times J$ is realized by merging x and y into the chain at those gaps, and at a common gap both relative orders of x, y occur, so $x \parallel y$ and the prescribed windows are exactly attained. $\square$

Lemma 2.2 (State bijection and counts). *Let $p = |I|$, $q = |J|$, $K = I \cap J$, $h = |K|$. The linear extensions of P are in bijection with the states $(i, j) \in I \times J$ with $i \neq j$, together with two ordered states for each common gap $i = j \in K$; hence*

$$N = e(P) = (pq - h) + 2h = pq + h.$$

The orientation counts of the incomparable pairs are

$$\begin{aligned}
D &= \#\{(i, j) \in I \times J : i < j\} + h && \text{for the pair } (x, y), \\
A_t &= q \#\{i \in I : i < t\} + \#\{i \in K : i < t\} && \text{for } (x, c_t), \ a < t \leq b, \\
B_t &= p \#\{j \in J : j < t\} + \#\{j \in K : j < t\} && \text{for } (y, c_t), \ c < t \leq d,
\end{aligned}$$

and exactly the displayed thresholds correspond to incomparable pairs; outside them the orientation is forced.

Proof. A linear extension is determined by the pair of gaps, except that when x and y occupy a common gap the two relative orders give two extensions — the doubled diagonal. For D : every off-diagonal state with $i < j$ contributes, and exactly the x -before- y diagonal state contributes, giving the displayed count. For A_t with $a < t \leq b$: $x < c_t$ exactly when the x -gap satisfies $i < t$; each such i pairs with q choices of j , plus one extra state when $i \in K$ (the doubled diagonal), and symmetrically for B_t . Incomparability of the displayed pairs is witnessed by explicit gap choices realizing both orientations. $\square$

3 The balance lemma

Theorem 3.1 (Overlapping Two-Interval Balance Lemma). *For every pair of nonempty overlapping windows, at least one of D , an eligible A_t , or an eligible B_t lies in $[N/3, 2N/3]$.*

The proof rests on one discrete intermediate-value fact and exact jump accounting.

Lemma 3.2 (Discrete crossing lemma). *Let $x_1 < \dots < x_r$ be integers with $x_1 \leq N/3$, $x_r \geq 2N/3$, and $x_{k+1} - x_k \leq N/3$ for all k . Then some $x_k \in [N/3, 2N/3]$.*

Proof. Take the first k with $x_k \geq N/3$. If $k = 1$ then $x_1 = N/3$ exactly and we are done; otherwise $x_{k-1} < N/3$, so $x_k = x_{k-1} + (x_k - x_{k-1}) < N/3 + N/3 = 2N/3$. $\square$

Exchanging the defect names replaces D by $N - D$ and swaps the A and B families; translation and order reversal also preserve the assertion. After these symmetries exactly two canonical forms remain: *crossing* ($a \leq c \leq b \leq d$; then $h = p - s$ where $s = c$ after translating a to 0, and $q \geq h$) and *nested* ($J \subseteq I$; then $h = q$).

The crossing form. Direct summation gives

$$N = pq + h, \quad D = pq - \frac{h(h-1)}{2}, \quad A_t = qt + \max(0, t - s), \quad B_{s+u} = pu + \min(u, h),$$

with exact consecutive jumps $A_{t+1} - A_t \in \{q, q+1\}$ and $B_{s+u+1} - B_{s+u} \in \{p, p+1\}$. If $N \geq 3(q+1)$ then $p \geq 2$, the A -sequence runs from $A_1 \leq q+1 \leq N/3$ to $A_{p-1} = N - (q+1) \geq 2N/3$ with jumps at most $q+1 \leq N/3$, and Lemma 3.2 produces a balanced eligible A_t . Symmetrically, if $N \geq 3(p+1)$ the B -sequence does the same. If neither large-case condition holds, the inequalities $q(p-3) < 3-h$ and $p(q-3) < 3-h$ with $1 \leq h \leq \min(p, q)$ force exactly fifteen triples (p, q, h) , each of which carries an explicit balanced witness:

(p, q, h)	s	N	witness
$(1, 1, 1)$	0	2	$D = 1$
$(1, 2, 1)$	0	3	$D = 2$
$(1, 3, 1)$	0	4	$B_{s+1} = 2$
$(1, 4, 1)$	0	5	$B_{s+1} = 2$
$(2, 1, 1)$	1	3	$D = 2$
$(2, 2, 1)$	1	5	$A_1 = 2$
$(2, 2, 2)$	0	6	$D = 3$
$(2, 3, 1)$	1	7	$A_1 = 3$
$(2, 3, 2)$	0	8	$D = 5$
$(3, 1, 1)$	2	4	$A_2 = 2$
$(3, 2, 1)$	2	7	$A_2 = 4$
$(3, 2, 2)$	1	8	$D = 5$
$(3, 3, 1)$	2	10	$A_2 = 6$
$(3, 3, 2)$	1	11	$A_2 = 7$
$(4, 1, 1)$	3	5	$A_2 = 2$

The nested form. Here $N = (p + 1)q$. If $q \geq 2$, take $u = \lceil q/3 \rceil$: then $1 \leq u \leq q - 1$ and $q \leq 3u \leq 2q$, so the eligible count $B_{s+u} = (p + 1)u$ satisfies $N \leq 3B_{s+u} \leq 2N$ *exactly* — the factorization of N makes the nested case rigid. If $q = 1$ then $N = p + 1$, $D = s + 1$, and $A_t = t + \mathbf{1}_{\{s < t\}}$; for $p \geq 5$ the A -sequence starts at most 2, ends at least $p - 1$, and jumps by 1 or 2, so Lemma 3.2 applies, while $p \leq 4$ is a finite table of explicit witnesses.

Remark 3.3 (D alone does not suffice). The count D alone is not always balanced: at $(p, q, h) = (2, 2, 1)$ one has $N = 5$ and $D = 4 > 2N/3$, while the eligible $A_1 = 2$ is balanced. Any proof must use the chain-cut counts.

4 The two-defect chain theorem

Theorem 4.1 (Two-Defect Chain Theorem). *Every finite non-chain poset P containing a chain C with $|P \setminus C| = 2$ has an incomparable pair (u, v) with $1/3 \leq \Pr(u \prec v) \leq 2/3$.*

Proof. Write $P \setminus C = \{x, y\}$. If x and y are comparable, then $C \cup \{x < y\}$ is a two-chain cover, so P has width at most two and the theorem of Linial [3] (in the sharpened form of Sah [16]) applies. If $x \parallel y$, their windows overlap by Lemma 2.1; Lemma 2.2 identifies the orientation counts of the genuine incomparable pairs, and Theorem 3.1 produces one in the middle third. $\square$

5 The extremal classification

Write Q for the three-element poset with exactly one relation. Aigner [5] proved that the width-two posets with balance constant exactly $1/3$ are precisely the ordinal sums of singletons and copies of Q . The census of Gupta [19] determines the extremal balance data through order 14: exactly 128 classes attain $1/3$, every one an ordinal sum of singletons and copies of Q — in particular all of width two. The following theorem proves this non-attainment for the

full (infinite, width-three-containing) class. Note the distinction between the *presentation* (a chain plus two designated defects, the model of Section 2, which any poset of defect at most two admits) and the *exact defect* $n - h(P)$: the extremal member Q arises as a one-element chain plus two defects but has exact defect one.

Theorem 5.1 (Equality rigidity). (a) *Every ordinal-sum-indecomposable poset of defect exactly two has an eligible orientation count strictly inside the open interval $(N/3, 2N/3)$; in particular its balance constant exceeds $1/3$.* (b) *Among ordinal-sum-indecomposable posets presented as a chain plus two incomparable defects, the unique member all of whose eligible counts avoid the open interval is Q — which has defect one. Every other member has an eligible count strictly inside.*

The proof classifies the *simultaneous avoiders* — parameter tuples for which every eligible count C satisfies $3C \leq N$ or $3C \geq 2N$ — and intersects them with an indecomposability criterion:

Proposition 5.2 (Avoider classification). *In the canonical crossing form, simultaneous avoidance holds exactly for $(p, q, h, s) = (1, 2, 1, 0)$ and $(2, 1, 1, 1)$. In the nested form, exactly for $(p, q, s) = (2, 1, 0)$ and $(2, 1, 1)$.*

Proposition 5.3 (Indecomposability criterion). *The two-defect model with windows $I = [a, b]$, $J = [c, d]$ is ordinal-sum-indecomposable iff its incomparability graph is connected, iff every $t \in \{1, \dots, m\}$ satisfies $a < t \leq b$ or $c < t \leq d$; under the overlap hypothesis this is equivalent to $\min(a, c) = 0$ and $\max(b, d) = m$.*

Proof of Proposition 5.2. Write $AV(C)$ for the condition $3C \leq N$ or $3C \geq 2N$. We first sharpen Lemma 3.2 under avoidance: if $x_1 \leq N/3$, $x_r \geq 2N/3$, all jumps are at most $N/3$, and every term satisfies AV , then the first term $\geq N/3$ equals $N/3$ exactly, and its successor equals $2N/3$ exactly (the predecessor of the first crossing term is strictly below $N/3$, so the jump bound puts that term strictly below $2N/3$; avoidance then forces equality at $N/3$, and the next term can neither be in the open middle nor overshoot $2N/3$).

Large regimes. In the crossing form the A -sequence has $A_1 \leq q + 1$, $A_{p-1} = N - (q + 1)$, and jumps in $\{q, q + 1\}$. If $N \geq 3(q + 1)$, all- A avoidance forces $N/3 \in \{q, q + 1\}$ by the sharpened lemma, hence $N = 3(q + 1)$, i.e. $q(p - 3) = 3 - h$. With $1 \leq h \leq \min(p, q)$ the solutions are $(p, q, h) = (3, q, 3)$ for $q \geq 3$, $(4, 2, 1)$, and $(5, 1, 1)$; the first is killed at $q = 3$ by $D = 6$, $N = 12$ (strict middle) and for $q > 3$ by the symmetric B -argument (which would force $N = 3(p + 1) = 12$, impossible), while the last two carry the strict witnesses $A_2 = 4$, $N = 9$ and $A_3 = 3$, $N = 6$. Symmetrically, $N \geq 3(p + 1)$ with all- B avoidance forces $p(q - 3) = 3 - h$, whose solutions $(p, 3, 3)_{p \geq 3}$, $(2, 4, 1)$, $(1, 5, 1)$ are killed the same way. Hence every survivor satisfies $N < 3(q + 1)$ and $N < 3(p + 1)$.

Residual list. The strict inequalities $q(p - 3) < 3 - h$ and $p(q - 3) < 3 - h$ force $p, q \leq 4$ and exactly the fifteen triples of the table in Section 3.1. All but two carry an eligible count strictly inside the open middle third:

$(1, 1, 1): D=1, N=2$	$(1, 3, 1): B_1=2, N=4$	$(1, 4, 1): B_1=2, N=5$
$(2, 2, 1): A_1=2, N=5$	$(2, 2, 2): D=3, N=6$	$(2, 3, 1): A_1=3, N=7$
$(2, 3, 2): D=5, N=8$	$(3, 1, 1): A_2=2, N=4$	$(3, 2, 1): A_2=4, N=7$
$(3, 2, 2): D=5, N=8$	$(3, 3, 1): A_2=6, N=10$	$(3, 3, 2): A_2=7, N=11$
$(4, 1, 1): A_2=2, N=5$		

(each displayed value C satisfies $N < 3C < 2N$). The remaining two triples avoid: at $(1, 2, 1)$, $N = 3$ with $D = B_1 = 2$, both at the boundary $3C = 2N$; at $(2, 1, 1)$, $N = 3$ with $D = 2$ and $A_1 = 1$, at the boundaries $3C = 2N$ and $3C = N$. Note $D - N/2 = (pq - h^2)/2 \geq 0$ throughout, so D never realizes the low branch in this orientation.

Nested form. Here $N = (p + 1)q$ and $B_{s+u} = (p + 1)u$. If $q \geq 2$ and $3 \nmid q$, then $u = \lceil q/3 \rceil$ gives $q < 3u < 2q$, a strict-middle B ; if $q = 3k \geq 6$, take $u = k + 1$. If $q = 3$: $p = 3$ gives $D = 6$, $N = 12$ strict middle, and for $p \geq 4$ the A -sequence starts at most $4 < N/3 = p + 1$, ends at least $3p - 1 > 2N/3$, with jumps at most 4, so Lemma 3.2 gives a strict term. For $q = 1$: $N = p + 1$ and $\{D\} \cup \{A_t\} = \{1, \dots, p\}$; $p = 1$ fails at $D = 1$, $N = 2$, $p \geq 3$ contains an integer strictly inside, and $p = 2$ survives for $s \in \{0, 1\}$ with values $\{1, 2\}$ at the boundaries of $N = 3$. $\square$

Proof of Proposition 5.3. The incomparability graph has the edge xy , no edges among chain vertices, and c_t adjacent to x exactly for $a < t \leq b$, to y exactly for $c < t \leq d$. After exchange assume $a \leq c$; overlap gives $c \leq b$, so the union of the two adjacency windows is the single index interval $(a, \max(b, d)]$, and the graph is connected exactly when that interval is all of $\{1, \dots, m\}$ — that is, $\min(a, c) = 0$ and $\max(b, d) = m$. Connectedness of the incomparability graph is equivalent to ordinal-sum indecomposability, since the canonical ordinal-sum factors are precisely the incomparability components. $\square$

Intersecting the avoider tuples (with their translations, defect exchanges, and order reversals) with the criterion leaves exactly the four interval presentations (L, F) , (F, L) , (F, R) , (R, F) , where $L = [0, 0]$, $R = [1, 1]$, $F = [0, 1]$ at $m = 1$: endpoint coverage forces $m = s + q - 1$, collapsing both survivor tuples to one-element chains with the displayed windows, each isomorphic to Q (extensions abc, acb, cab ; pair counts $(2, 1)$ and $(1, 2)$). Every other arithmetic survivor has a nontrivial chain prefix or suffix and is a chain- Q -chain ordinal sum, hence decomposable. This proves part (b) of Theorem 5.1.

Part (a) follows. Let P be indecomposable of defect exactly two, and let C be a maximum chain, $P \setminus C = \{x, y\}$. If x and y are comparable, $C \cup \{x < y\}$ is a two-chain cover, so P has width two; Linial [3] gives $\delta(P) \geq 1/3$, and by Aigner's classification [5] equality holds only for ordinal sums of singletons and copies of Q — all decomposable except Q itself, whose defect is one. Hence $\delta(P) > 1/3$: some incomparable pair lies strictly inside. If $x \parallel y$, part (b) applies to the presentation, and its unique avoider Q has defect one, not two; so P carries a strict eligible count.

Remark 5.4. An eligible count with $N/3 < C < 2N/3$ yields a strict witness; if all eligible counts avoid the open interval, Theorem 3.1 forces equality at $N/3$ or $2N/3$ — which is the situation classified above. A corroborating exact scan through $p, q \leq 60$ (73,810 canonical cases) found no further avoiders.

6 The next rung: three defects

The machinery of this paper is the $k = 2$ instance of a general weighted-gap model: for k defects, linear extensions correspond to feasible gap assignments weighted by the extension counts of the tied blocks. The first three-defect result is already in reach of a symmetry argument, and it lands exactly on the conjecture's boundary.

Theorem 6.1 (Aligned Q -defect chains). *Let $P \setminus C = \{x, y, z\}$ induce $x < y$ with z incomparable to both, all three insertion windows equal to a common interval I with $p = |I| \geq 2$. Then $N = e(P) = p(p+1)(p+2)/2$ and*

$$\Pr(x \prec z) = \Pr(z \prec y) = 2/3 \quad \text{exactly.}$$

Proof sketch. States are triples of gaps (i, j, k) with $i \leq j$, weighted 1, 2, or 3 by how many defects share a gap. Delete the relation $x < y$: the three defects become interchangeable and relabeling splits the extensions into six classes of equal size $\binom{p+2}{3}$. Restoring $x < y$ retains exactly the classes xyz, xzy, zxy ; two of the three have $x \prec z$ and two have $z \prec y$. $\square$

Thus the aligned family generalizes the extremal poset Q of Theorem 5.1 to every window size, with its defect pairs pinned to the boundary. Beyond the aligned chamber the two-defect arithmetic genuinely stops: with three defects, tied fibers are no longer pair-additive (a fully tied gap carries the extension count of the induced three-element defect order — 6, 3, 2, or 1 — rather than a product of pairwise corrections), and orientation counts become asymmetric between the defect pairs. Deciding the full three-defect analogue of Theorem 4.1, and the extremal classification above width three, are the natural next problems; the interval-order connection (incomparable-defect k -defect chains are interval orders with k nontrivial intervals) suggests the ladder climbs toward genuinely open territory.

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